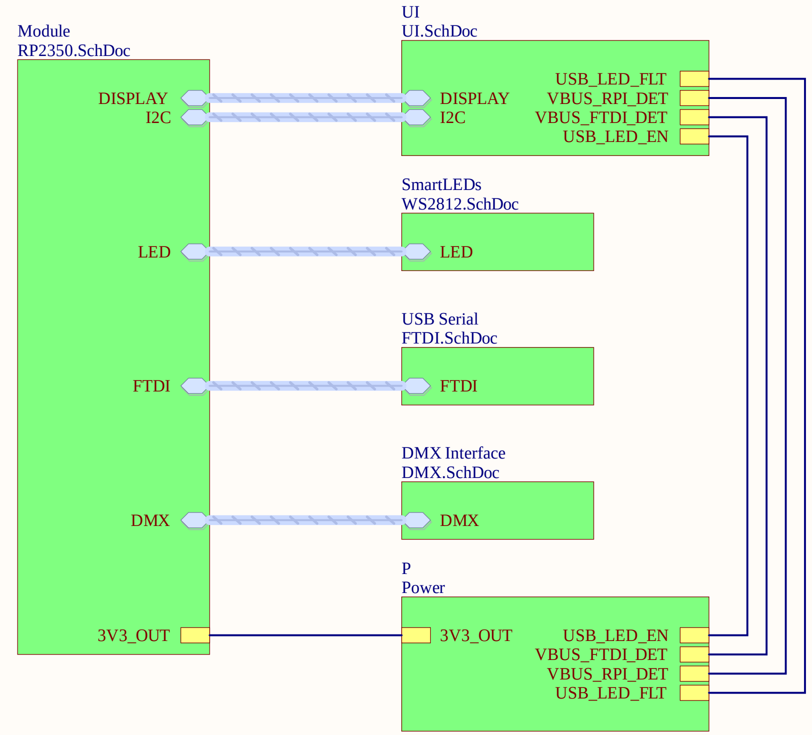
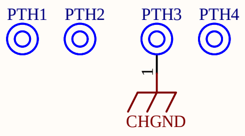




AC/DC Power Supply

5V 200W (194.0mm x 55.0mm x 20.0mm)
https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200A-5/7707187

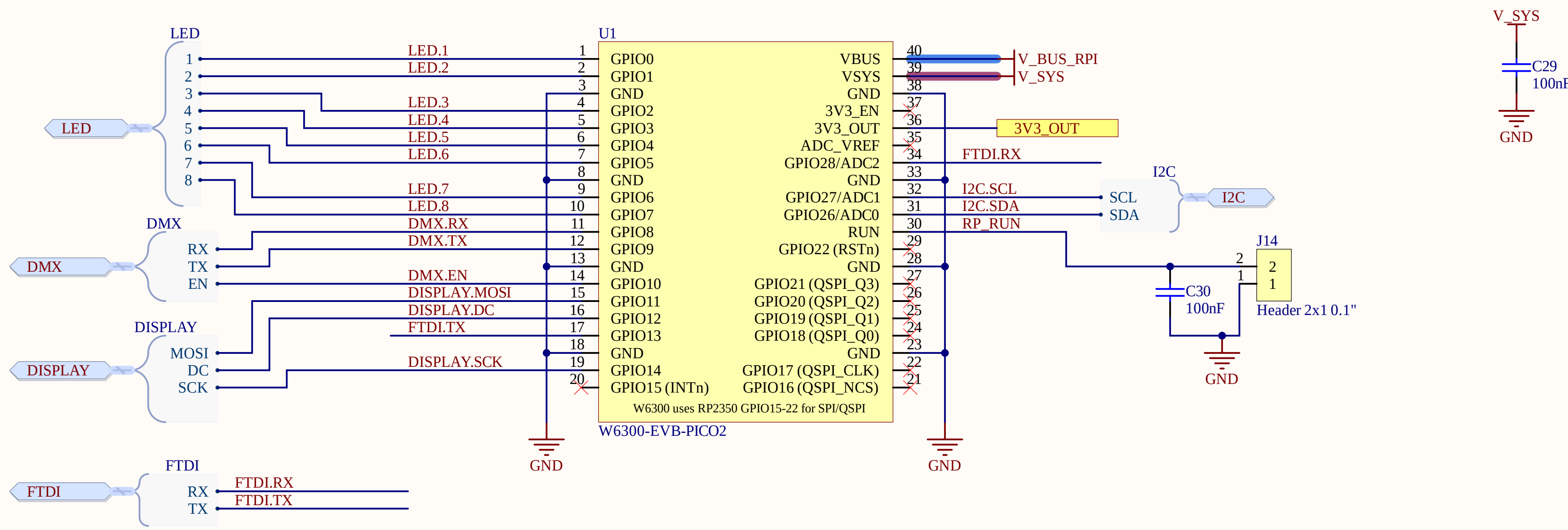
12V 162W (194.0mm x 55.0mm x 20.0mm)
https://www.digikey.com/en/products/detail/mean-well-usa-inc/LSP-160-12T/11689253

12V 200W (194.0mm x 55.0mm x 20.0mm)
https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200-12/7707237

24V 162W (194.0mm x 55.0mm x 20.0mm)
https://www.digikey.com/en/products/detail/mean-well-usa-inc/LSP-160-24T/11689249

24V 200W (194.0mm x 55.0mm x 20.0mm)
https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200-24/7707239

Title: Block Diagram		
Size: B	Number: DMX Interface	Revision: 2
Date: 9/04/2026		Sheet of 1 of 7
File: C:\Users\..\Block Diagram.SchDoc		Drawn By: *

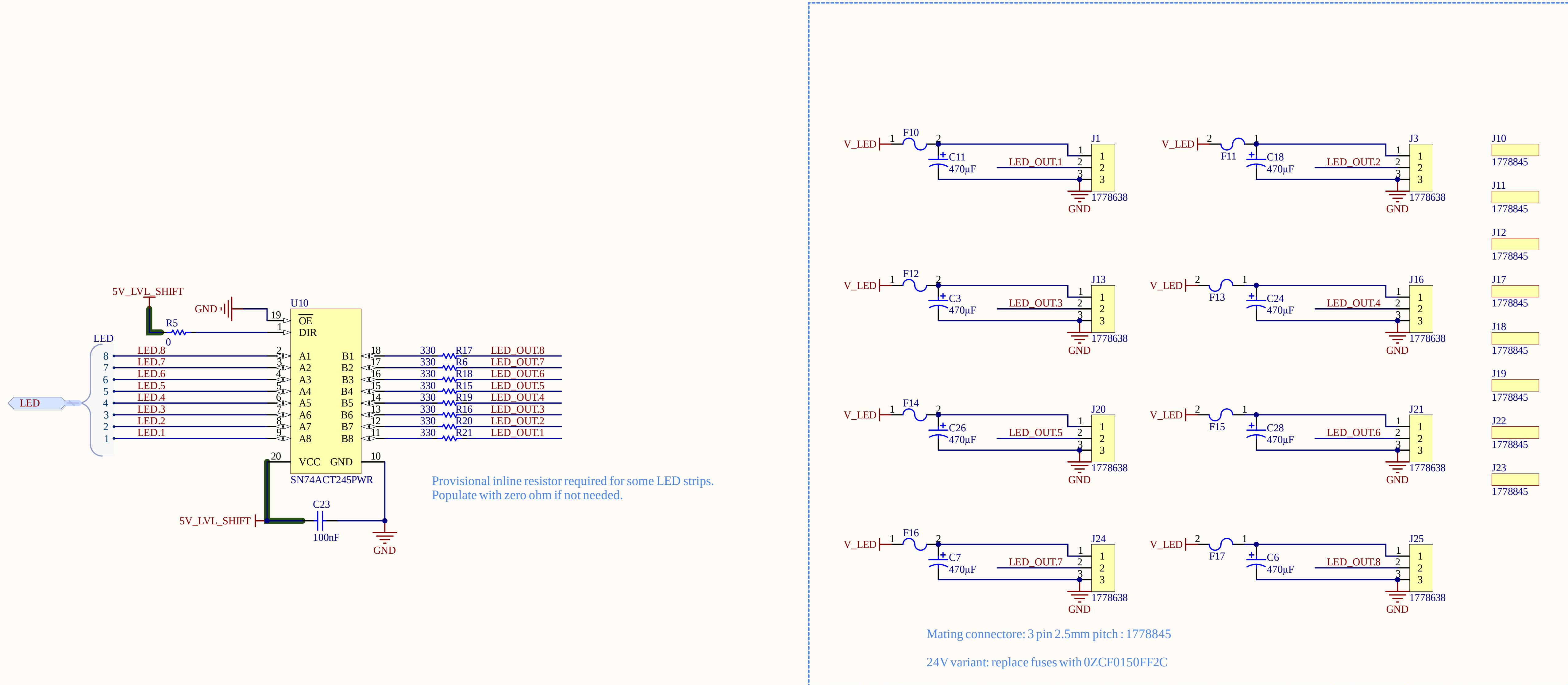


Socket the module on two 1x20 sockets

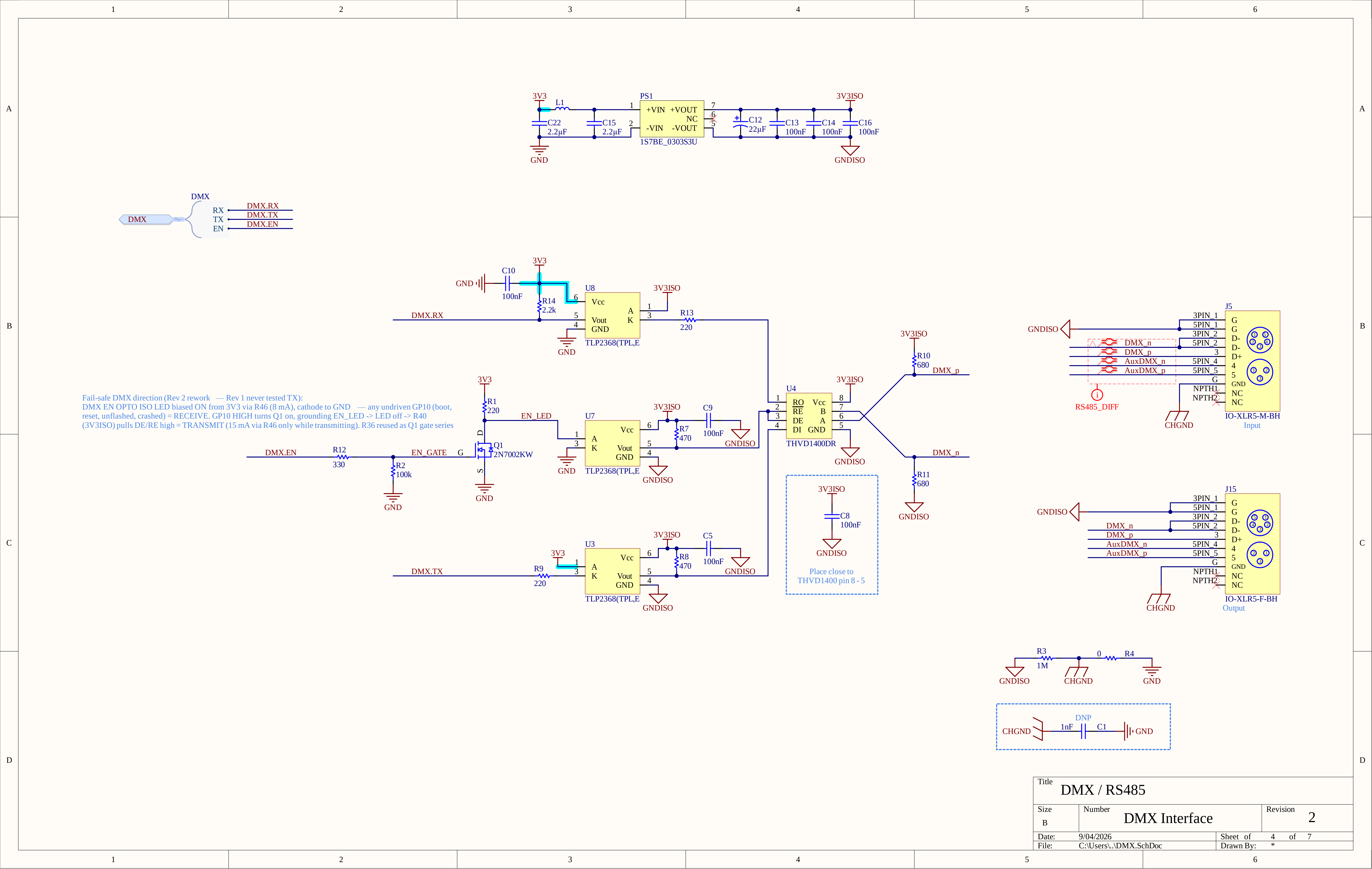
GP15-22 (pins 20-22, 24-27, 29) are hard-wired to the on-module W6300; they are marked no-connect and must stay that way.

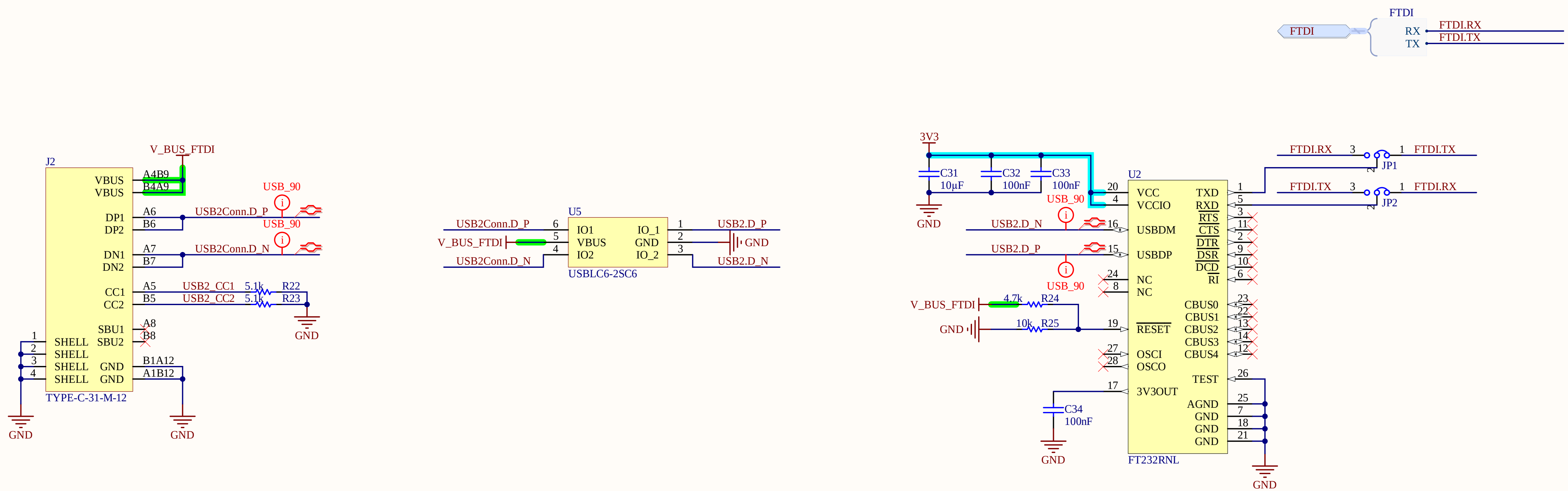
3V3_OUT (pin 36) feeds ONLY the carrier LDO - the sequencing gate that keeps carrier 3V3 from rising before IOVDD. No carrier load runs from it.

Title: RP2350			
Size: B	Number: DMX Interface		Revision: 2
Date: 9/04/2026		Sheet 2 of 7	
File: C:\Users\..\RP2350.SchDoc		Drawn By: *	



Title: Smart LED		
Size: B	Number: DMX Interface	Revision: 2
Date: 9/04/2026		Sheet 3 of 7
File: C:\Users\..\WS2812.SchDoc		Drawn By: *


3V3
L1
PS1
+VIN +VOUT
NC
-VIN -VOUT
1S7BE_0303S3U
C22 2.2µF
C15 2.2µF
C12 22µF
C13 100nF
C14 100nF
C16 100nF
3V3ISO
GND
GNDISO
DMX
RX
TX
EN
DMX.RX
DMX.TX
DMX.EN
C10 100nF
R14 2.2k
U8
Vcc
Vout
GND
A
K
TLP2368(TPL,E
R13 220
Fail-safe DMX direction (Rev 2 rework — Rev 1 never tested TX):
DMX EN OPTO ISO LED biased ON from 3V3 via R46 (8 mA), cathode to GND — any undriven GP10 (boot, reset, unflashed, crashed) = RECEIVE. GP10 HIGH turns Q1 on, grounding EN_LED -> LED off -> R40 (3V3ISO) pulls DE/RE high = TRANSMIT (15 mA via R46 only while transmitting). R36 reused as Q1 gate series
R1 220
EN_LED
EN_GATE
R12 330
R2 100k
Q1 2N7002KW
U7
R7 470
C9 100nF
U4
RO
RE
DE
DI
Vcc
B
A
GND
THVD1400DR
R10 680
R11 680
DMX_p
DMX_n
3V3ISO
C8 100nF
GNDISO
Place close to THVD1400 pin 8 - 5
U3
R9 220
R8 470
C5 100nF
J5
3PIN_1
5PIN_1
3PIN_2
5PIN_2
3
5PIN_4
5PIN_5
G
NPTH1
NPTH2
D-
D+
4
5
NC
DMX_n
DMX_p
AuxDMX_n
AuxDMX_p
RS485_DIFF
CHGND
IO-XLR5-M-BH
Input
J15
IO-XLR5-F-BH
Output
R3 1M
R4 0
DNP
1nF C1
Title DMX / RS485
Size B
Number DMX Interface
Revision 2
Date: 9/04/2026
Sheet 4 of 7
File: C:\Users\..\DMX.SchDoc
Drawn By: *

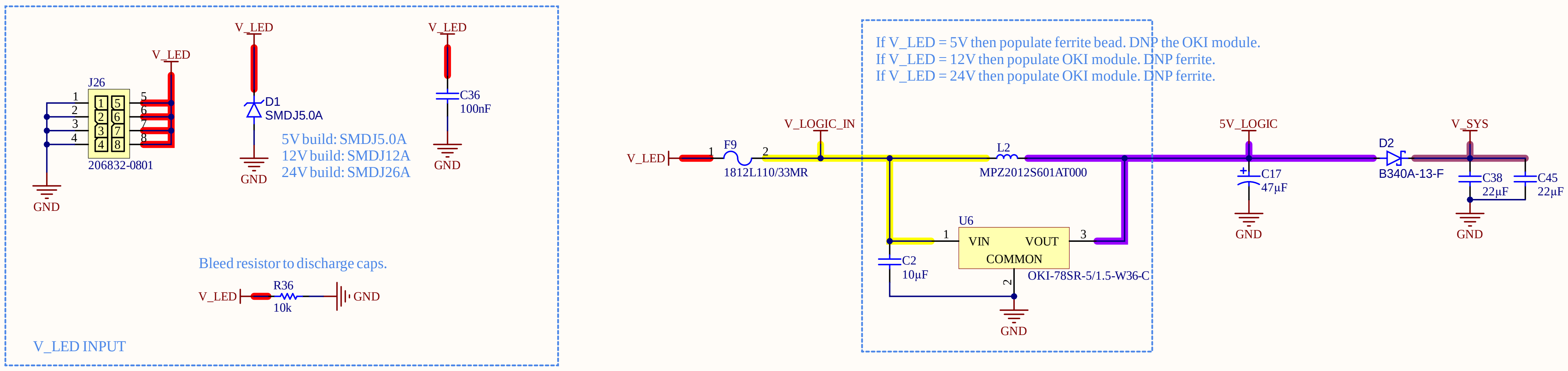


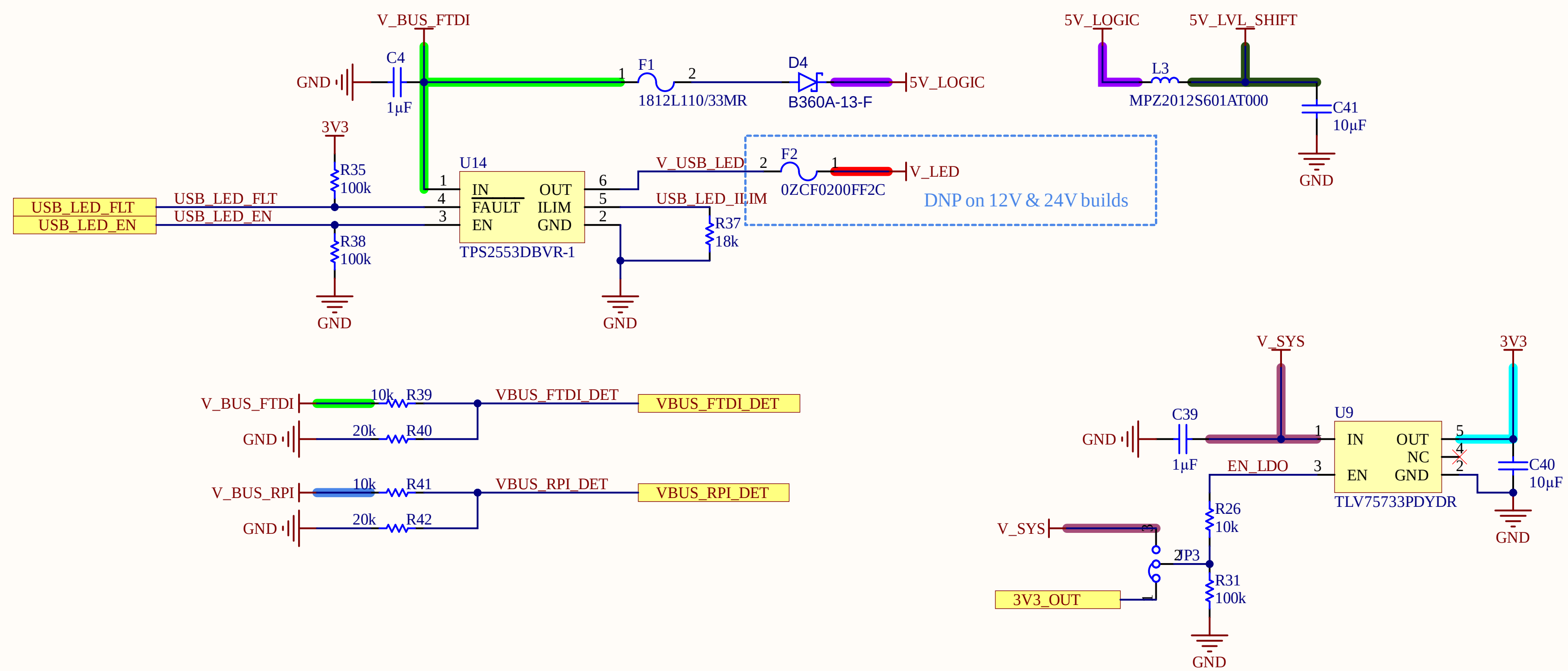
This FT232R is self-powered from the board's always-on 3V3, unlike a normal bus-powered dongle. A running FT232R asserts its internal 1.5 kΩ pull-up on D+ (that's how USB devices announce themselves). If RESET# were simply tied high, then whenever the box is on, the chip would drive ~3.3 V into D+ of whatever it's cabled to — telling a sleeping PC "a device just attached," back-feeding an unpowered hub port, and generally misbehaving on the far end. Held in reset, D+ stays released until real VBUS proves a host is actually there. This is verbatim FTDI's recommended self-powered configuration from the FT232R datasheet (their §"Self Powered Configuration" shows this exact 4.7 k/10 k divider), and it's why the schematic note on the usb_dmx sheet reads "no host, no enumeration, no back-drive."

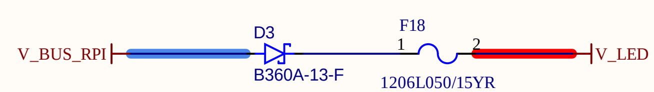
The contrast, for completeness: in a bus-powered design (chip's VCC comes from VBUS), you'd just tie RESET# to VCC or leave it floating on its internal pull-up, because the whole chip dies with the cable anyway. Ours doesn't — hence the gate.

One practical bench note: this means the FT232R port only enumerates when both the box is powered and a host is attached — if you ever probe USB2_VBUS at 5 V but FTDI_RST isn't ~3.4 V, check RESET line voltage divider resistors stuffing before anything else.

Title			
Size B	Number		Revision
Date:	9/04/2026	Sheet of	
File:	C:\Users\..\FTDI.SchDoc	Drawn By:	







INPUT: NO main fuse on the board: a 40-50 A MIDI/ANL fuse in the supply harness at the PSU is REQUIRED — it protects the V_LED plane and is what D1 crowbars against on overvoltage. Per-port protection: 8 x PTC fuses on the WS2812 sheet; F9 guards logic.

V_LED is the strip rail: 5 V OR 12 V per BUILD (one voltage per board — silk-mark it).

3V3 here IS the diagram net 3V3_CARRIER: TLV75733 (dropout 450 mV max at 1 A, ~80 mV at our ~175 mA incl. FT232RNL + TFT backlight) regulates from 3.95 V worst-case USB input. Thermal pad to GND pour. Module 3V3(OUT) feeds nothing on the carrier.

SEQUENCING: U1 EN high = on, FAULT to P07, 18k ~ 1.45A, is gated by the module 3V3_OUT, so the carrier 3V3 rail rises only after the RP2350 IOVDD is up — without this, the I2C pull-ups and opto anodes back-inject ~15 mA into unpowered, non-failsafe RP2350 pads at every power-up.

F2 DNP on 12/24 V, PSU trim ≥ 5.2 V.

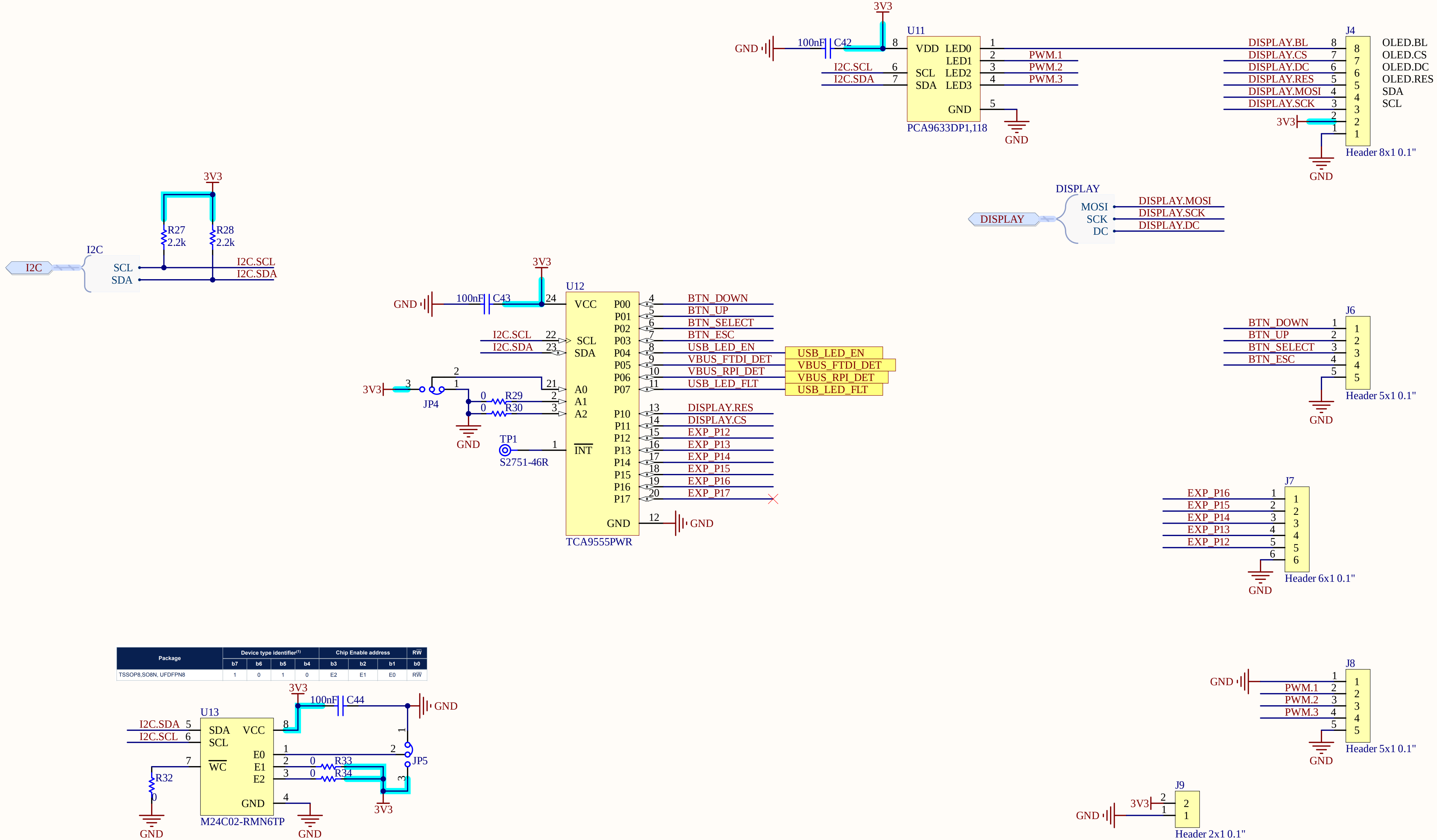
RATING: 8 ports x 5 A = 40 A worst-case through the board

Layout: 2 oz copper, V_LED and GND as paired pours, strip returns meet logic ground at J1 only.

CHGND bond: R4 fitted / C1 DNP are the tuning points.

THERMAL: D1 (B340A, SMA ~90 K/W on decent pads) dissipates ~200 mW at the ~550 mA logic+TFT load (~+20 C) — give it copper on both pads. VSYS = 5.0 V - Vf ~ 4.6 V, ample for the module buck. Same SMA pad takes onsemi SS34 or MCC SK34.

Title		
Size: B	Number	Revision
Date: 9/04/2026		Sheet of
File: C:\Users\..\Power.SchDoc		Drawn By:



Package	Device type identifier(1)				Chip Enable address			RW̅
	b7	b6	b5	b4	b3	b2	b1	b0
TSSOP8,SO8N, UFDFPN8	1	0	1	0	E2	E1	E0	RW̅

7-bit I2C address 0x50 throgh 0x57
Default to 0x56

Title		
Size: B	Number	Revision
Date: 9/04/2026		Sheet of
File: C:\Users\..\UI.SchDoc		Drawn By: